



**NORTH  
SYDNEY  
GIRLS HIGH  
SCHOOL**

**2020**

**Preliminary  
Assessment  
Task 2**

# Preliminary Mathematics Extension

## General Instructions

- Reading Time – 2 minutes
- Working Time – 55 minutes
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided
- In Questions 6 – 8, show relevant mathematical reasoning and/or calculations

**Total marks – 41**

**Section I – 5 marks** (pages 2 – 3)

- Attempt Questions 1 – 5
- Allow about 7 minutes for this section

**Section II – 36 marks** (pages 4 – 6)

- Attempt Questions 6 – 8
- Allow about 48 minutes for this section

**NAME:** \_\_\_\_\_

**TEACHER:** \_\_\_\_\_

| Question                      | 1 – 5 | 6   | 7   | 8   | Total |
|-------------------------------|-------|-----|-----|-----|-------|
| Further Algebra and Functions | /2    | /5  | /13 |     | /20   |
| Further Trig                  | /3    | /7  |     | /11 | /21   |
|                               | /5    | /12 | /13 | /11 | /41   |

## Section I

5 marks

Attempt Questions 1- 5

Allow about 7 minutes for this section

Use the multiple-choice answer sheet for Questions 1- 5

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1. Which expression is equivalent to  $\cos(x + 80^\circ)$  ?

(A)  $\sin x \cos 80^\circ + \cos x \cos 80^\circ$

(B)  $\sin x \cos 80^\circ - \cos x \cos 80^\circ$

(C)  $\cos x \cos 80^\circ + \sin x \sin 80^\circ$

(D)  $\cos x \cos 80^\circ - \sin x \sin 80^\circ$

2. If  $\sin \alpha = \frac{5}{13}$ , what are the possible values for  $\sin 2\alpha$  ?

(A)  $\frac{10}{13}$

(B)  $\pm \frac{10}{13}$

(C)  $\frac{120}{169}$

(D)  $\pm \frac{120}{169}$

3. Which of the following describes the range of the function  $f(x) = 5 \sin x + 12 \cos x$  ?

(A)  $f(x) \leq 13$

(B)  $f(x) \leq 17$

(C)  $|f(x)| \leq 13$

(D)  $|f(x)| \leq 17$

4. Which of the following inequations is NOT equivalent to the inequation  $|3x - 1| > 4$  ?

(A)  $|1 - 3x| > 4$

(B)  $3x - 1 > \pm 4$

(C)  $(3x - 1)^2 > 16$

(D)  $\frac{1}{|3x - 1|} < \frac{1}{4}$

5. Consider the parametric equations:

$$x = \frac{1 - t^2}{1 + t^2}$$

$$y = \frac{2t}{1 + t^2}$$

These equations describe:

(A) a straight line

(B) a parabola

(C) a circle

(D) a hyperbola

## Section II

**36 marks**

**Attempt Questions 6-8**

**Allow about 48 minutes for this section**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In questions 6-8, your responses should include relevant mathematical reasoning and/or calculations.

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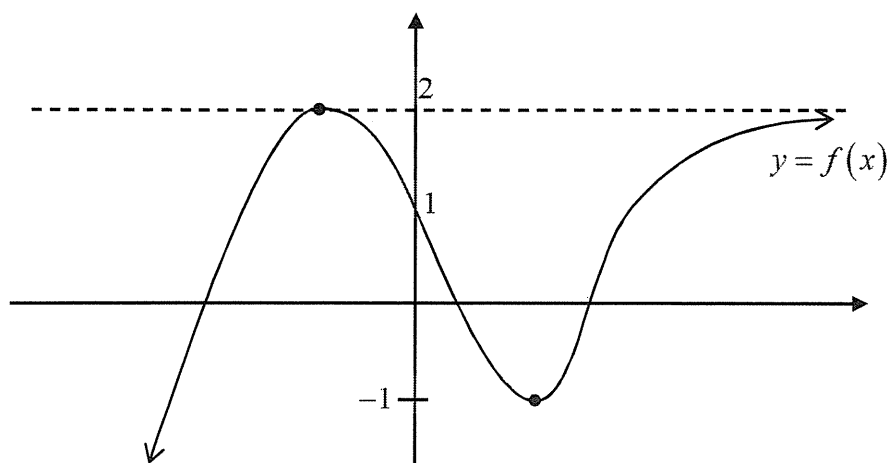
**Question 6** (12 marks) Use a SEPARATE writing booklet.

- (a) Solve for  $x$ :  $x^3 - 4x \leq 0$  2
- (b) (i) Find, in fully expanded form, the Cartesian equation of the function with parametric equations: 2
- $$x = 2t - 1$$
- $$y = 4t^2 + 1$$
- (ii) Find the range of this function using interval notation, given that  $t$  is restricted to the interval  $(-2, 1)$ . 1
- (c) Find a general solution to the equation  $\cos 2\theta = \frac{\sqrt{3}}{2}$ . 2
- (d) (i) By writing  $\cos 3x = \cos(2x + x)$ , show that  $\cos 3x = 4\cos^3 x - 3\cos x$ . 3
- (ii) Hence fully simplify  $8\cos^3\left(\frac{\pi}{6} - x\right) - 6\cos\left(\frac{\pi}{6} - x\right)$ . 2

**End of Question 6**

**Question 7** (13 marks) Use a SEPARATE writing booklet.

- (a) Consider the following graph of the function  $y = f(x)$  :



On the Graph Response Sheet supplied, sketch the graphs of the following relations, showing all important information:

- (i)  $y = \frac{1}{f(x)}$  2
- (ii)  $y = f(|x|)$  2
- (b) Solve for  $x$ :  $\frac{x}{x+2} \leq 2$  3
- (c) (i) By using the technique of addition of ordinates, or otherwise, sketch the graph of  $y = 2^x + 2^{-x}$ , showing the coordinates of any important points. 2
- (ii) Write down the largest domain containing  $x = -1$  for which the inverse of this function is a function. 1
- (iii) Show that the equation in part (i) can be written:  $2^{2x} - y(2^x) + 1 = 0$  1
- (iv) Hence find the equation of the inverse function specified by part (ii). 2

**End of Question 7**

**Question 8** (11 marks) Use a SEPARATE writing booklet.

- (a) Use the product-to-sum results to show that

2

$$4 \cos 4x \cos 2x \cos x = \cos 7x + \cos 5x + \cos 3x + \cos x.$$

- (b) Let  $7 \cos \theta - 4 \sin \theta = R \cos(\theta + \alpha)$ .

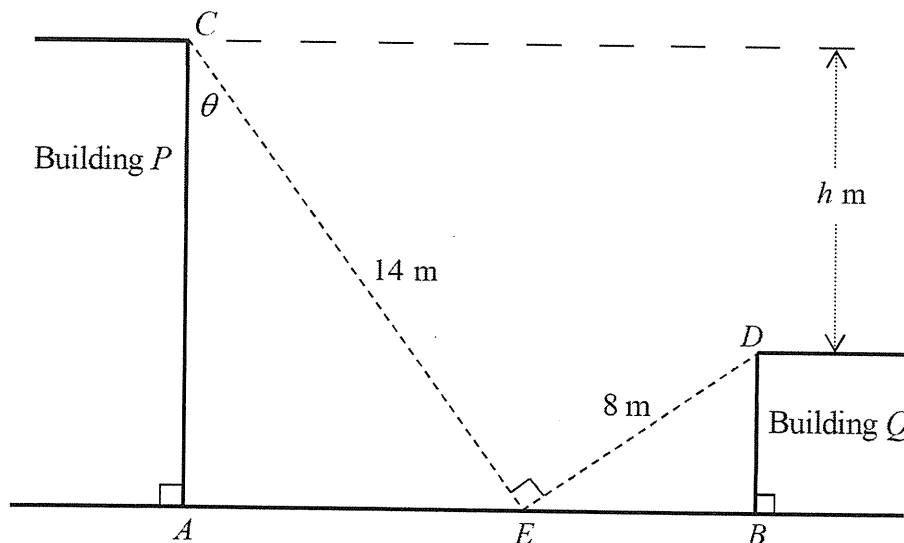
- (i) By expanding the right-hand side and equating coefficients, show that  $R = \sqrt{65}$  and  $\alpha = 29^\circ 45'$  (to the nearest minute).

3

- (ii) Hence solve  $7 \cos \theta = 4 \sin \theta + 2$  for  $0^\circ \leq \theta \leq 360^\circ$ .

2

- (c) The diagram shows two buildings on opposite sides of a street, one building  $h$  metres taller than the other. A rope of length 14 metres is joined to the top  $C$  of the taller building at one end, and to a point  $E$  on the road at the other end. Another rope of length 8 metres is tied to the top  $D$  of the smaller building at one end, and to the same point  $E$  at the other. It is found that  $EC \perp ED$ .  $A$ ,  $B$ ,  $C$ ,  $D$  and  $E$  all lie in the same plane. Let  $\angle ACE = \theta$ .



- (i) Explain why  $\angle BED = \theta$ .
- (ii) Form an expression involving  $\theta$  (and no other pronumerals) for  $h$ , the difference in the heights of the buildings.
- (iii) If building  $P$  is 4 metres taller than building  $Q$ , use your expression from part (ii) and the result of part (b) to find the height of the taller building.

1

1

2

**End of Examination**

## 2020 Yr 11 2U Test 2 Solutions

Section I Summary: 1. D 2. D 3. C 4. B 5. C

Solutions

$$\begin{array}{l|l} 2. \sin 2\alpha = 2\sin\alpha\cos\alpha & \cos^2\alpha = 1 - \sin^2\alpha \\ & = 1 - \left(\frac{5}{13}\right)^2 \\ & = \frac{144}{169} \\ & \cos\alpha = \pm \frac{12}{13} \\ & \sin\alpha = \pm \frac{5}{13} \end{array}$$

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3. Using aux. angle method,  $5\sin x + 12\cos x = R\sin(x+\alpha)$   
where  $R = \sqrt{5^2 + 12^2} = 13$   
 $\therefore -\sqrt{13} \leq y \leq \sqrt{13}$

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4. (A)  $|-x| = |x|$  so  $|1-3x| = |3x-1|$  ✓

(C)  $\sqrt{x^2} = |x|$  so  $|3x-1| > 4$   
 $\sqrt{(3x-1)^2} > 4$   
 $(3x-1)^2 > 16$  ✓

(D) Provided  $c > 0$ , if  $x > c$  then  $0 < \frac{1}{x} < \frac{1}{c}$  ✓

(B) x correct solution:  $3x-1 < -4$ ,  $3x+1 > 4$

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5.  $x = \cos\theta$  given  $t = \tan\frac{\theta}{2}$   
 $y = \sin\theta$

These are parametric equations of a circle:

$$\begin{aligned} x^2 + y^2 &= \cos^2\theta + \sin^2\theta \\ &= 1 \end{aligned}$$

## Question 6

- (a) Solve for  $x$ :  $x^3 - 4x \leq 0$

2

$$x(x^2 - 4) \leq 0$$

$$x(x-2)(x+2) \leq 0$$



$$x \leq -2, 0 \leq x \leq 2$$

$$\text{OR } x \in (-\infty, -2] \cup [0, 2]$$

- (b) (i) Find, in fully expanded form, the Cartesian equation of the function with parametric equations:

2

$$x = 2t - 1$$

$$y = 4t^2 + 1$$

$$x = 2t - 1$$

$$2t = x + 1$$

$$t = \frac{x+1}{2}$$

$$y = 4t^2 + 1$$

$$y = 4\left(\frac{x+1}{2}\right)^2 + 1$$

$$y = 4 \cdot \frac{(x+1)^2}{4} + 1$$

$$y = (x+1)^2 + 1$$

$$y = x^2 + 2x + 2$$

- (ii) Find the range of this function using interval notation, given that  $t$  is restricted to the interval  $(-2, 1)$ .

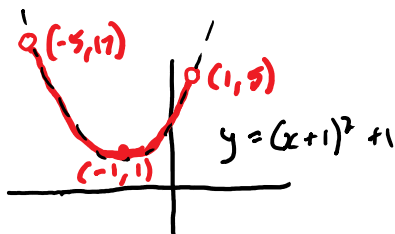
1

$$t = -2 \Rightarrow x = -5$$

$$y = 17$$

$$t = 1 \Rightarrow x = 1$$

$$y = 5$$



$$\text{Range: } 1 \leq y < 17$$

$$\text{OR } y \in [1, 17)$$

- (c) Find a general solution to the equation  $\cos 2\theta = \frac{\sqrt{3}}{2}$ .

2

$$2\theta = \frac{\pi}{6} + 2k\pi, \frac{11\pi}{6} + 2k\pi \quad \text{where } k \text{ is an integer}$$

$$\theta = \frac{\pi}{12} + k\pi, \frac{11\pi}{12} + k\pi$$

- (d) (i) By writing  $\cos 3x = \cos(2x+x)$ , show that  $\cos 3x = 4\cos^3 x - 3\cos x$ .

3

$$\begin{aligned}\cos 3x &= \cos(2x+x) \\ &= \cos 2x \cos x - \sin 2x \sin x \\ &= (\cos^2 x - \sin^2 x) \cos x - 2 \sin x \cos x \cdot \sin x \\ &= \cos^3 x - 3 \sin^2 x \cos x \\ &= \cos^3 x - 3(1 - \cos^2 x) \cos x \\ &= \cos^3 x - 3 \cos x + 3 \cos^3 x \\ &= 4 \cos^3 x - 3 \cos x\end{aligned}$$

- (ii) Hence fully simplify  $8\cos^3\left(\frac{\pi}{6}-x\right) - 6\cos\left(\frac{\pi}{6}-x\right)$ .

2

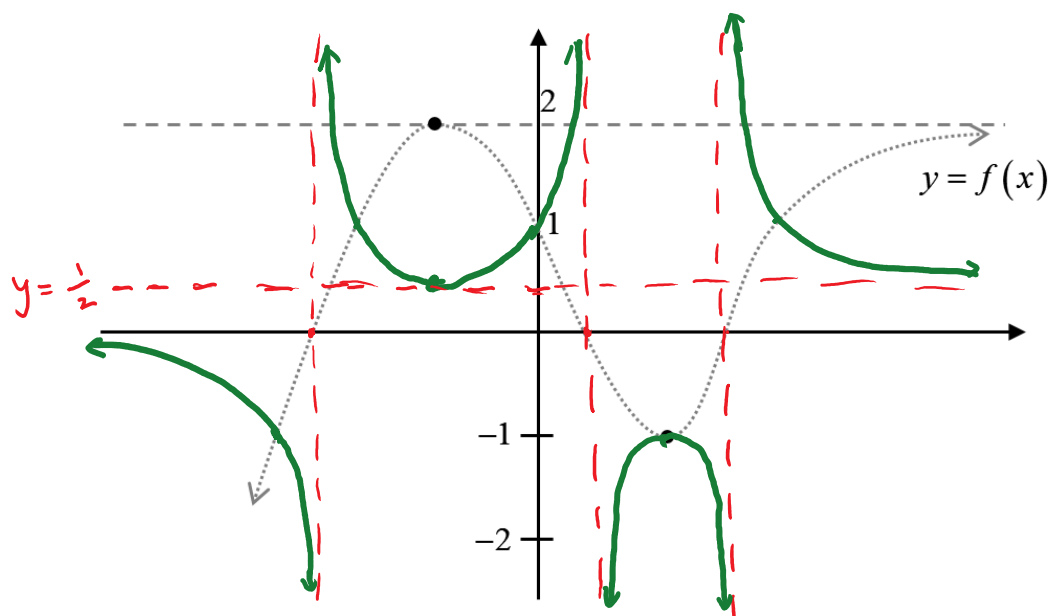
$$\begin{aligned}&8\cos^3\left(\frac{\pi}{6}-x\right) - 6\cos\left(\frac{\pi}{6}-x\right) \\ &= 2\left[4\cos^3\left(\frac{\pi}{6}-x\right) - 3\cos\left(\frac{\pi}{6}-x\right)\right] \\ &= 2\cos\left[3\left(\frac{\pi}{6}-x\right)\right] \quad (\text{from part i}) \\ &= 2\cos\left(\frac{\pi}{2}-3x\right) \\ &= 2\sin 3x\end{aligned}$$

# Question 7

(a)

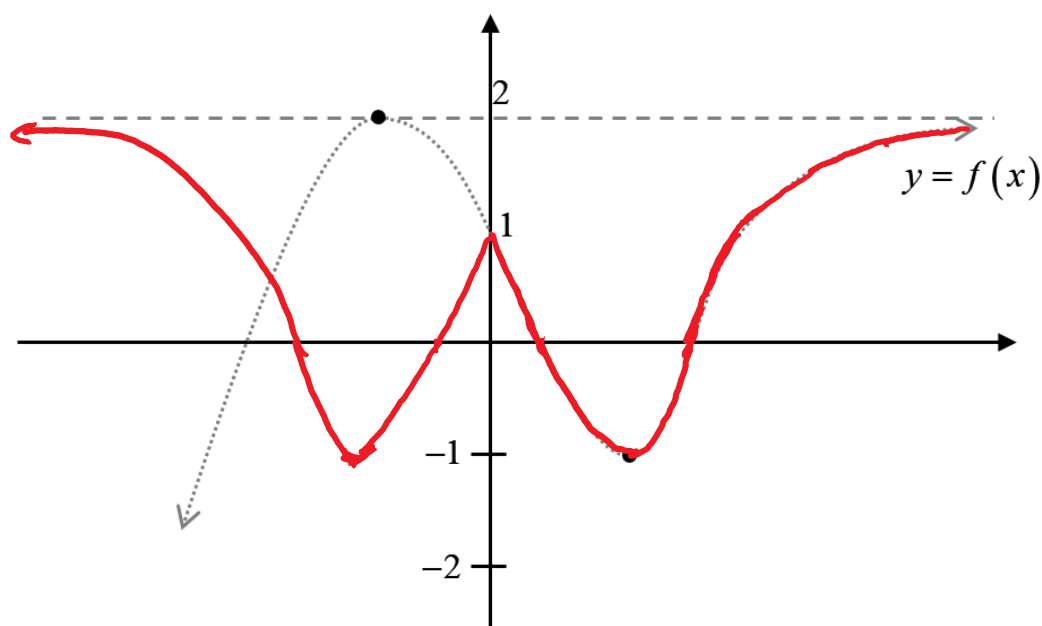
- (i) Sketch  $y = \frac{1}{f(x)}$ .

2



- (ii) Sketch  $y = f(|x|)$ .

2



(b) Solve for  $x$ :  $\frac{x}{x+2} \leq 2$

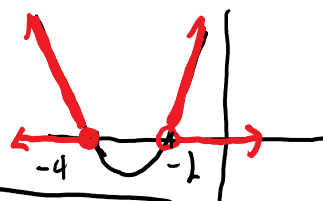
3

$$x(x+2)^2: x(x+2) \leq 2(x+2)^2, x \neq -2$$

$$2(x+2)^2 - x(x+2) > 0$$

$$(x+2)[2(x+2) - x] > 0$$

$$(x+2)(x+4) > 0, x \neq -2$$

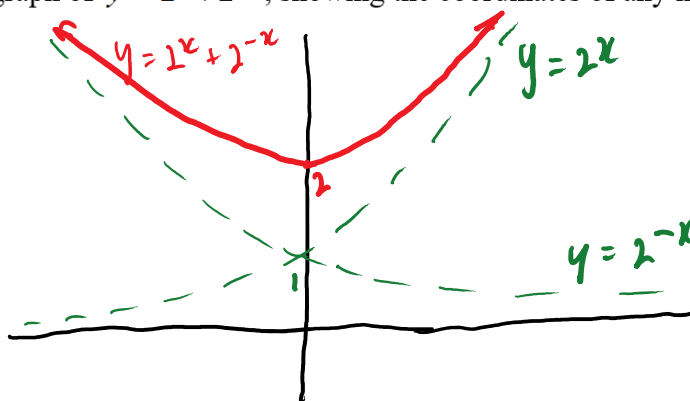


$$\boxed{x \leq -4 \vee x > -2}$$

$$\text{OR } x \in (-\infty, -4] \cup (-2, \infty)$$

- (c) (i) By using the technique of addition of ordinates, or otherwise, sketch the graph of  $y = 2^x + 2^{-x}$ , showing the coordinates of any important points.

2



- (ii) Write down the largest domain containing  $x = -1$  for which the inverse of this function is a function.

1

$$x \leq 0 \quad \text{OR} \quad x \in (-\infty, 0]$$

- (iii) Show that the equation in part (i) can be written:  $2^{2x} - y(2^x) + 1 = 0$

1

$$y = 2^x + 2^{-x}$$

$$\times 2^x: y(2^x) = 2^{2x} + 1$$

$$2^{2x} - y(2^x) + 1 = 0$$

(iv) Hence find the equation of the inverse function specified by part (ii).

2

$$2^{2x} - y(2^x) + 1 = 0$$

$$2^x = \frac{y \pm \sqrt{y^2 - 4}}{2} \quad (\text{quadratic formula})$$

$$\text{for specified domain: } 2^x = \frac{y - \sqrt{y^2 - 4}}{2}$$

$$x = \log_2 \left[ \frac{y - \sqrt{y^2 - 4}}{2} \right]$$

$$\therefore \text{Inverse: } y = \log_2 \left[ \frac{y - \sqrt{y^2 - 4}}{2} \right]$$

$$\text{which may be written: } y = \log_2 (y - \sqrt{y^2 - 4}) - 1$$

$$4 \cos 4x \cos 2x \cos x = \cos 7x + \cos 5x + \cos 3x + \cos x.$$

$$\begin{aligned} 4 \cos 4x \cos 2x \cos x &= 2 \cos 4x \cos 2x \cdot 2 \cos x \\ &= \left[ \cos(4x + 2x) + \cos(4x - 2x) \right] \cdot 2 \cos x \\ &= 2 \cos 6x \cos x + 2 \cos 2x \cos x \\ &= \cos(6x + x) + \cos(6x - x) + \cos(2x + x) + \cos(2x - x) \\ &= \cos 7x + \cos 5x + \cos 3x + \cos x \end{aligned}$$

(b) Let  $7 \cos \theta - 4 \sin \theta = R \cos(\theta + \alpha)$ .

(i) By expanding the right-hand side and equating coefficients, show that  $R = \sqrt{65}$  and  $\alpha = 29^\circ 45'$  (to the nearest minute).

3

$$\begin{aligned} 7 \cos \theta - 4 \sin \theta &= R \cos(\theta + \alpha) \\ &= R(\cos \theta \cos \alpha - \sin \theta \sin \alpha) \\ &= (R \cos \alpha) \cos \theta - (R \sin \alpha) \sin \theta \end{aligned}$$

Equating coefficients of  $\cos \theta$  &  $\sin \theta$ :

$$R \cos \alpha = 7 \quad \text{--- (1)}$$

$$R \sin \alpha = 4 \quad \text{--- (2)}$$

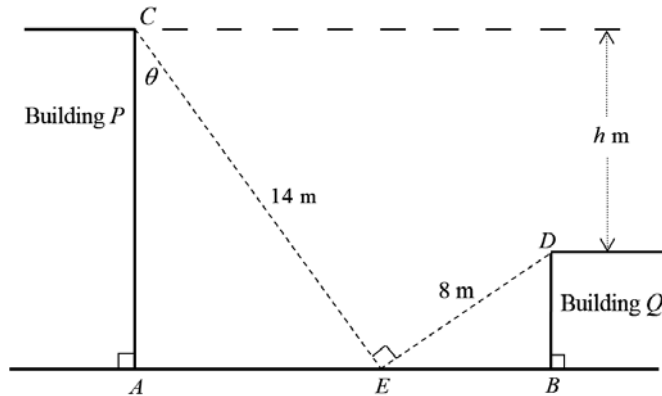
$$\begin{aligned} \textcircled{1}^2 + \textcircled{2}^2: \quad R^2 \cos^2 \alpha + R^2 \sin^2 \alpha &= 7^2 + 4^2 \quad \textcircled{1} \div \textcircled{1}: \quad \tan \alpha = \frac{4}{7} \\ R^2 (\cos^2 \alpha + \sin^2 \alpha) &= 65 \\ R^2 &= 65 \\ R &= \sqrt{65} \end{aligned} \quad \left| \begin{array}{l} (\alpha \text{ in 1st quadrant as } \sin \alpha > 0, \cos \alpha > 0) \\ \alpha = \tan^{-1} \frac{4}{7} \\ = 29^\circ 45' \end{array} \right.$$

(ii) Hence solve  $7 \cos \theta = 4 \sin \theta + 2$  for  $0^\circ \leq \theta \leq 360^\circ$ .

2

$$\begin{aligned} 7 \cos \theta - 4 \sin \theta &= 2 \\ \sqrt{65} \cos(\theta + 29^\circ 45') &= 2 \\ \cos(\theta + 29^\circ 45') &= \frac{2}{\sqrt{65}} \\ \theta + 29^\circ 45' &= \cos^{-1} \frac{2}{\sqrt{65}}, 360 - \cos^{-1} \frac{2}{\sqrt{65}} \\ &= 75.64^\circ, 284.36^\circ \\ \theta &= 45^\circ 53', 254^\circ 37' \end{aligned}$$

- (c) The diagram shows two buildings on opposite sides of a street, one building  $h$  metres taller than the other. A rope of length 14 metres is joined to the top  $C$  of the taller building at one end, and to a point  $E$  on the road at the other end. Another rope of length 8 metres is tied to the top  $D$  of the smaller building at one end, and to the same point  $E$  at the other. It is found that  $EC \perp ED$ .  $A, B, C, D$  and  $E$  all lie in the same plane. Let  $\angle ACE = \theta$ .



- (i) Explain why  $\angle BED = \theta$ .

1

$$\begin{aligned}\hat{AEC} &= 180 - 90 - \hat{ACE} \quad (\text{angle sum of } \triangle) \\ &= 90 - \theta \\ \hat{BED} + \hat{DEC} + \hat{AEC} &= 180^\circ \quad (\text{angles on straight line}) \\ \hat{BED} + 90 + 90 - \theta &= 180 \\ \hat{BED} &= \theta\end{aligned}$$

- (ii) Form an expression involving  $\theta$  (and no other pronumerals) for  $h$ , the difference in the heights of the buildings.

1

$$\begin{aligned}h &= AC - DB \\ &= 14 \cos \theta - 8 \sin \theta\end{aligned}$$

- (iii) If building  $P$  is 4 metres taller than building  $Q$ , use your expression from part (ii) and the result of part (b) to find the height of the taller building.

2

$$\begin{aligned}h &= 4 \\ 14 \cos \theta - 8 \sin \theta &= 4 \\ 7 \cos \theta - 4 \sin \theta &= 2 \\ \text{From (b)(ii), } \theta &= 45^\circ 53', 254^\circ 37' \\ \text{but } \theta < 90^\circ, \text{ so } \theta &= 45^\circ 53' \\ \text{height of taller building} &= 14 \cos 45^\circ 53' \\ &= 9.7 \text{ m}\end{aligned}$$